

→ 4-  $A \times \emptyset = \emptyset$  , prove.

$$|A| \cdot |\emptyset| = |A| \cdot 0 = 0 \quad , \quad \checkmark$$

$$\rightarrow \underline{|A \times \emptyset|} = |A| \cdot |\emptyset|$$
$$|A| \cdot 0 = \underline{0}$$

→ we see that car... of  $A \times \emptyset$   
equals to 0.

→ since <sup>only</sup> the cardinality of  $\emptyset$  is  
0, thus

$$A \times \emptyset = \emptyset$$

9- (d)  $\mathcal{D}(\underline{\{ \emptyset, 1 \}}, \underline{\{ \emptyset, 1 \}})$

$$\{ \emptyset, \{ \emptyset, 1 \}, \{ \emptyset, 1, \{ \emptyset, 1 \} \} \}$$

~~$\{ \emptyset, 1 \}$~~  no duplicate